

Table 3: Mean number of visual entities and in-scene text references present per video, partitioned on video type. Same 210 video subset is used for analysis as that used for the analysis shown in Table 1.

	News coverage			Edited footage			Raw footage		
	Vis.	Text	Total	Vis.	Text	Total	Vis.	Text	Total
Disaster ("What")	1.42	2.38	3.80	1.14	0.41	1.55	1.05	0.17	1.22
Place ("Where")	-	2.51	2.51	-	0.59	0.59	-	0.39	0.39
Time ("When")	-	1.26	1.26	-	0.41	0.41	-	0.16	0.16
Affected people ("Who")	1.48	0.94	2.42	1.18	0.23	1.41	0.86	0.04	0.90
First responders ("Who")	1.73	0.78	2.51	1.36	0.45	1.81	0.17	0.12	0.29
Disaster outcomes	1.28	-	1.28	0.77	0.27	1.04	0.67	-	0.67
Total	5.91	7.87	13.78	4.45	2.36	6.81	2.75	0.88	3.63



Figure 3: Snapshot of video news coverage of the 2019 Notre Dame fire news cycle (an event in MultiVENT). The fire and the fallen spire were initially reported through first-person social media video uploads (at 10:02 AM and 10:51 AM, respectively) and then later broadcast by news organizations in more detail (10:38 AM, 11:10 AM, 12:20 PM, 12:26 PM). Some news coverage (12:20 PM) directly used first-person social media footage (10:51 AM). Hours later, news agencies uploaded more complete news stories with details and context (6:20 PM). This data suggests that it is important for models to learn from both first-person videos and official news coverage at various points in the news cycle to fully construct a factual model of the event, especially if the model is attempting to construct an event model while information develops online.

provided with the full set of videos V and a text query $t \in T$, and for each video $v \in V$ returns the probability that t describes v , or $h(v, t) = \mathbb{P}[f(v, t) = 1]$. When there is a bijective mapping between queries and videos (e.g., when using video descriptions as queries), the model is evaluated on its recall when considering the top 1, 5, and 10 ranked videos ($R@1$, $R@5$, and $R@10$), as well as the median rank (MedR). When a given query may describe multiple videos, (e.g., when using event descriptions as queries), we instead evaluate the model on its precision given the top 1, 5, and 10 ranked videos ($P@1$, $P@5$, and $P@10$). We define these metrics as:

$$\text{Given } S := \arg \max_{V' \subseteq V: |V'|=k} \sum_{v \in V'} h(v, t),$$

Table 4: MultiCLIP evaluated alongside existing video retrieval approaches on the video retrieval benchmark MSR-VTT. Results indicate that MultiCLIP performs adequately on existing retrieval tasks, achieving comparable results to existing models. It does not perform as well as architectures that use multimodal transformers for joint encodings such as InternVideo and MPLUG-2.

Method	Year	Rank@1	Rank@5	Rank@10
FrozenInTime [4]	2021	32.5	61.5	71.2
Clip2Video [16]	2021	29.8	55.5	66.2
InternVideo [47]	2022	55.2	79.6	87.5
MPLUG-2 [50]	2023	53.1	77.6	84.7
MultiCLIP	2023	38.4	70.1	82.7

$$R@k = \frac{|\{s \in S : f(s, t) = 1\}|}{|\{v \in V : f(v, t) = 1\}|} \quad \text{and} \quad P@k = \frac{|\{s \in S : f(s, t) = 1\}|}{k}.$$

5.2 Model architecture and training

We introduce MultiCLIP, a multilingual baseline for video retrieval on MultiVENT. We base our architecture on the pretrained LAION CLIP ViT-H/14 frozen XLM-Roberta-Large model [11], which jointly trains an image and text encoder on text-image data to learn to pair images with their captions. At test time, it produces a zero-shot linear layer based on the test input’s visual features through which natural language captions can be passed in. The model architecture contains a vision encoder based on a ViT architecture [14] and a text encoder based on the the multilingual XLM Roberta large model [12]. A full overview of the CLIP architecture and pretraining can be found in the original paper [33].

In experiments using MultiCLIP, we first tokenize text descriptions using the XLM-Roberta-Large tokenizer, containing a vocabulary of over 250,000 words, and pass the tokens into MultiCLIP which produces a text embedding of size 1024. Next, we uniformly sample videos at a rate of 12 frames per video with an input size of 224x224, which the model uses to create a frame embedding of size 1024. To incorporate multilinguality into the model’s frame-level features, we use a ViT architecture trained with a contrastive objective over multilingual image-caption pairs from the LAION-5B dataset [42], which is constructed from the Common Crawl archive using images and their alt-text to produce a multilingual image-text dataset with over 100 languages. We mean pool the frame embeddings to produce a final video embedding, and use the text and video features to compute a similarity matrix of videos and descriptions.

5.3 Retrieval baselines

We first evaluate MultiCLIP on the existing video retrieval task MSR-VTT [51] using the recall metrics described in Sec. 5.1 alongside contemporary video retrieval approaches (FrozenInTime [4], Clip2Video [16], InternVideo [47], and MPLUG-2 [50]). Results on MSR-VTT’s validation set are reported in Table 4. The results indicate MultiCLIP performs well on standard video retrieval tasks, matching performance of separate text/vision pipeline models released within the last two years. It performs better than existing models that use separate text and vision pipelines (FrozenInTime [4] and Clip2Video [16]), but not as well as models that use larger architectures involving multimodal encodings (InternVideo [47] and MPLUG-2 [50]).

5.4 MultiVENT retrieval

We now evaluate MultiCLIP and related retrieval approaches on MultiVENT. We first use multilingual video descriptions as queries, and then we use English event summaries taken from the grounding text documents, meaning that one text query maps to up to ten videos. The event queries are selected by taking one to two sentences from each English event text document that describes the event most holistically. We exclusively use English queries for this section, as our annotators fluent in the other languages were not available for this task. In addition to MultiCLIP, we consider a set of contemporary video retrieval models with lightweight architectures (FrozenInTime [4], CLIP2Video

Table 5: Results showing the retrieval performance of video retrieval methods alongside MultiCLIP on MultiVENT. We use video descriptions and event descriptions as queries and partition results based on language. As shown, MultiVENT can be a difficult retrieval benchmark for video retrieval models even when considering only English, but the benefit of training on multilingual data is apparent when comparing MultiCLIP against the regular pooled CLIP model on non-English data.

Method	Video description				Event description		
	R@1	R@5	R@10	MedR	P@1	P@5	P@10
English							
FrozenInTime [4]	6.5	20.0	28.4	53.0	42.3	34.6	26.9
CLIP2Video [16]	41.3	71.8	80.4	2.0	96.2	96.9	73.3
InternVideo [47]	53.8	83.1	88.7	1.0	94.2	93.5	79.6
CLIP (pooled) [33]	55.9	83.9	91.3	1.0	98.1	94.6	80.6
MultiCLIP	55.9	84.5	92.3	1.0	100.0	96.9	80.6
Arabic + Chinese + Korean + Russian							
FrozenInTime [4]	0.5	1.2	2.5	793.5	29.8	22.6	17.6
CLIP2Video [16]	2.4	7.2	10.5	166.5	14.4	9.4	7.3
InternVideo [47]	5.7	13.9	19.8	91.0	79.3	71.0	55.7
CLIP (pooled) [33]	6.2	15.9	22.4	79.5	83.7	73.3	58.2
MultiCLIP	32.6	64.7	79.5	3.0	85.6	76.4	61.5

[16], InternVideo [47], and a pooled CLIP model using the same setup as MultiCLIP without the additional multilingual pretraining). We argue that lightweight architectures are most appropriate for evaluating a full, pairwise set of similarity scores between text and video data of large multimodal corpora. Results are reported, partitioned on language, in Table 5.

We report the standard recall @ rank k metric for retrieval on individual video queries, and precision @ rank k for retrieval on event description queries. The results suggest that some existing video retrieval models may particularly struggle on this task, regardless of language. We hypothesize that this is due to a combination of the videos’ length, complex semantic content, ambiguity, and frequent OCR content, as well as the long and often noisy video description queries.

While MultiVENT as a whole poses challenges to existing models, it is also clear that multilingual data may significantly impact performance on models trained primarily on English content - all models suffer a performance loss when evaluated on multilingual content (even when using English queries, as shown by the event description query results). While MultiCLIP suffers a performance loss on this data as well, comparing the standard pooled CLIP model against MultiCLIP shows that training on multilingual data does mitigate this multilingual performance loss: The two models perform comparably on English data, but MultiCLIP performs better on the multilingual content, especially when multilingual queries are used.

6 Conclusion

We introduce MultiVENT, a multimodal, multilingual dataset grounded in natural language documents for event-centric video retrieval and information acquisition. This dataset consists of 2,396 videos covering 260 current events reported in five target languages (Arabic, Chinese, English, Korean, and Russian) paired with multilingual natural language video descriptions and long-form event-centric text documents. We use this dataset to characterize online news coverage and how models can use this online content for information acquisition. We propose a multilingual video retrieval benchmark using MultiVENT and present MultiCLIP, multilingual video retrieval model to serve as a baseline for the task. We evaluate this model and related retrieval approaches on MSR-VTT and MultiVENT to illustrate the importance of pretraining on multilingual data for evaluation on MultiVENT. In future work, we aim to explore the effect that joint vision-OCR embeddings can have on video retrieval in text-heavy contexts. Also in future work, a RePAQ-adjacent system [25] for automatically extracting question-answer pairs from video content and video-document pairs could be developed and applied to MultiVENT. Through this, a framework for teaching models to perform open-domain question-answering tasks with multimodal background corpora could be established, expanding the domain of questions a model can answer.